

Model Assessment (Evaluation/Validation) and Model Selection (part I)

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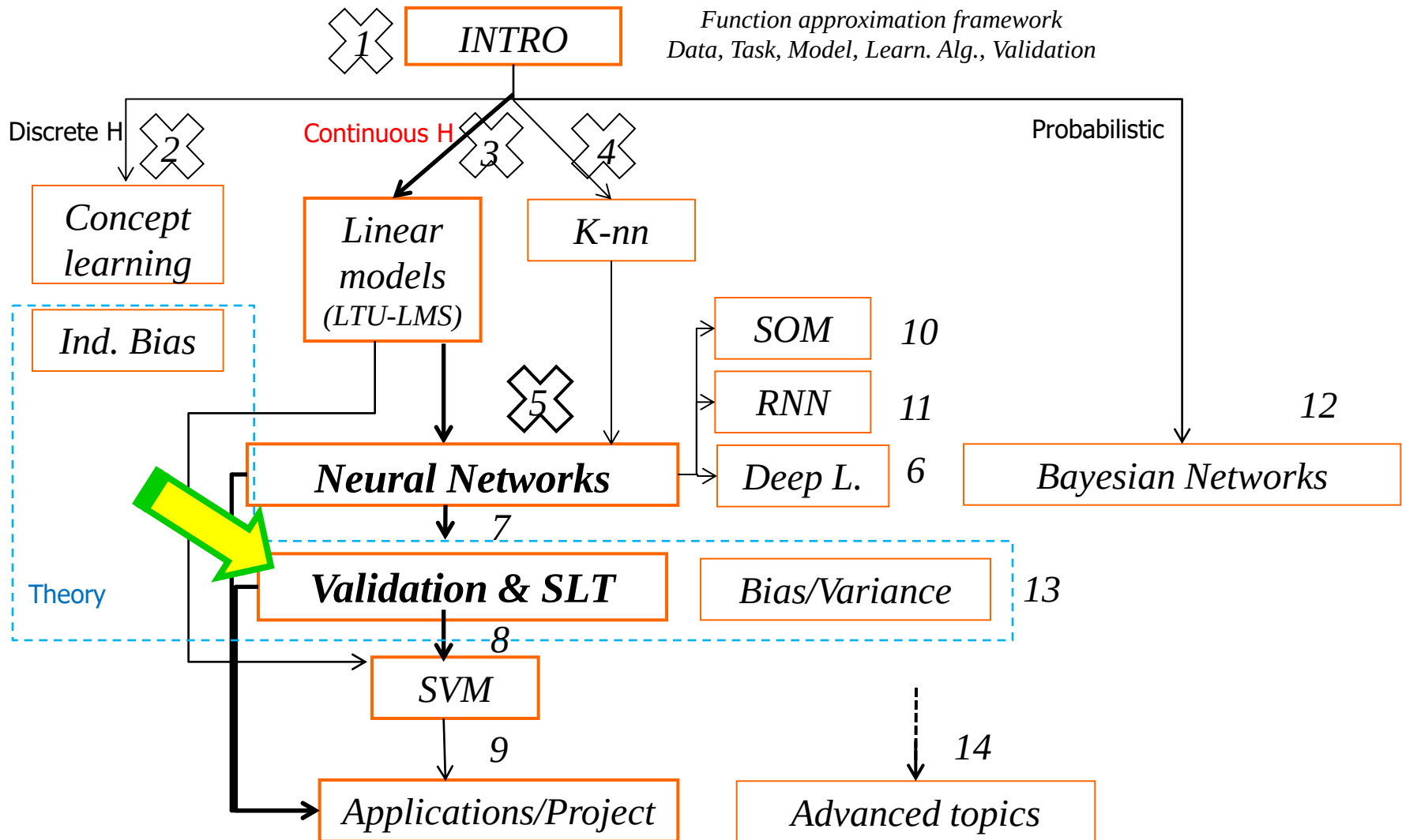
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ML Course structure

Where we go



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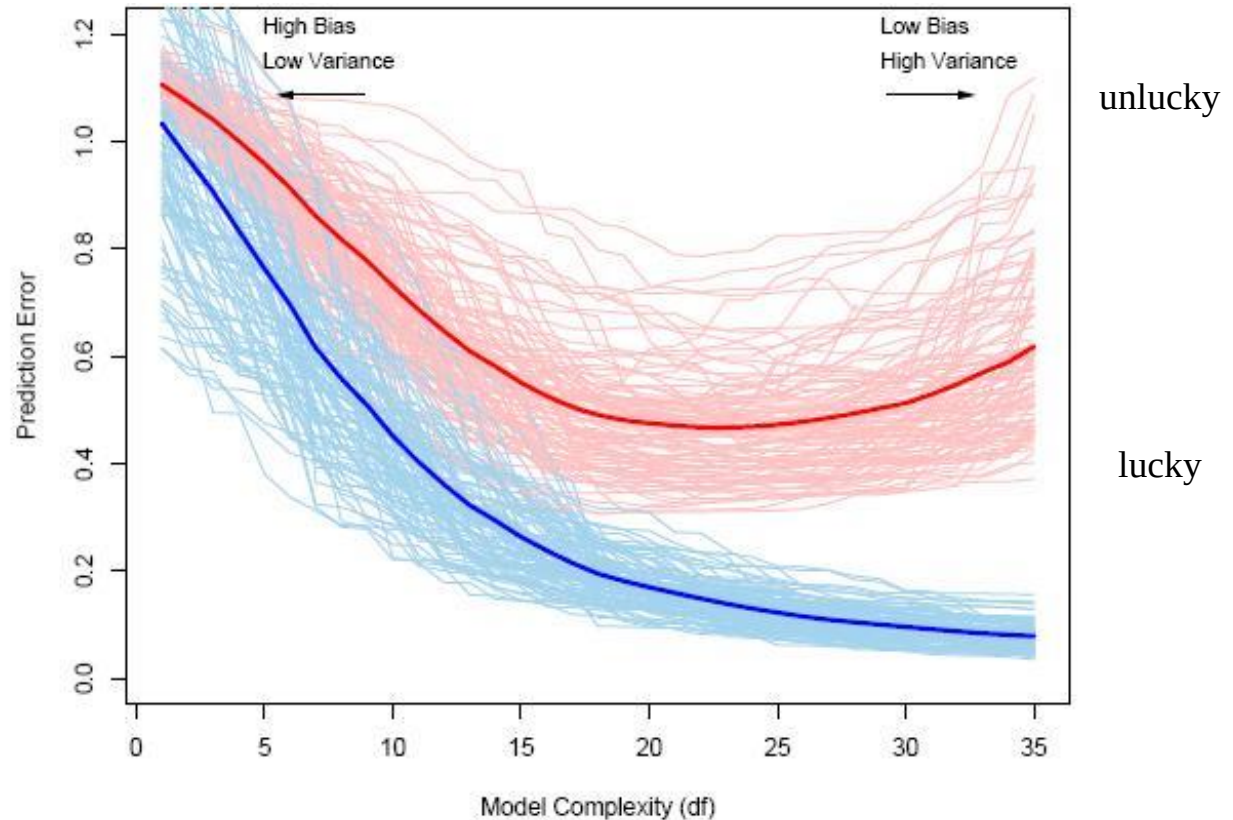
A premise: Bias-Variance

- Bias-Variance decomposition provides an useful framework to understand the validation issue, showing how the estimation of a model performance *is difficult*
 - Considering also the role of different training set realizations
 - Showing again the need of a trade-off between fitting capability (bias) and model flexibility (variance) in different way
- Anyway we will discuss it later by a specify lecture
- Let me introduce just this plot, to recall the need to optimize a model in the right way.

Under & Overfitting in a Bias-Variance plot*



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Complex model : lower
bias, higher variance →
possible overfitting

FIGURE 7.1. Behavior of test sample and training sample error as the model complexity is varied. The light blue curves show the training error $\overline{\text{err}}$, while the light red curves show the conditional test error $\text{Err}_{\mathcal{T}}$ for 100 training sets of size 50 each, as the model complexity is increased. The solid curves show the expected test error Err and the expected training error $E[\overline{\text{err}}]$.

Motivations

- Recall that we are looking for the best solution (minimal prediction – test error) searching a **balancing** between **fitting (accuracy on training data)** and **model complexity**
 - Training set **is not** a good estimate of test error
 - Initially ok but you have too many bias (underfitting, high TR error) than too high variance (low TR error but overfitting)
- Assuming a tuning hyper-parameters θ (implicit or explicit) that varies the model complexity, we wish to find the value of θ that minimizes test error
 - Look for methods for estimating the expected error for a model (or each model of a class of models or even of a set of models*)

Approximating estimation step (intro: a guide)

We can approximate the estimation

- Analytically:
 - AIC, BIC Akaike/Bayesian Information Criterion
 - limited to linear models (in par.)
 - MDL (Minimum Description Length)
 - SRM: **Structural Risk Minimization & VC-dimension** →
 - In lecture INTRO + a next lecture on the VC-DIM

In practice, often we can approximate the estimation on the data set:

- By resampling: direct estimation of extra-sample error via
 - cross-validation (hold-out, K-fold CV, ...)
 - bootstrap

Validation: Two aims

Recall the slides in the intro (on the validation)! **THE MOST IMPORTANT SLIDE** of the course

- After models training on the training set
- **Model selection:** estimating the performance (*generalization error*) of different learning models in order to choose the best one (to generalize)
 - this includes search the best *hyper-parameters* of your model (e.g. polynomial order, #units in a NN, lambda, eta,).

It returns a model

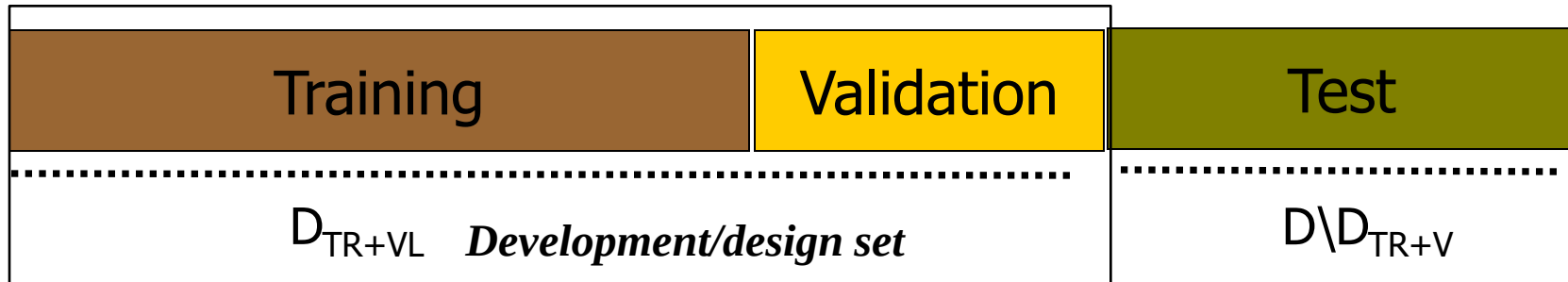
- **Model assessment:** having chosen a final model (or a class of models), estimating/evaluating its prediction error/ risk (*generalization error*) on new test data (measure of the quality of the ultimately chosen model)

It returns an estimation value

Gold rule: Keep separation between goals and use separate data sets in each phase

Hold out

- If data set size is sufficient: e.g. 50% TR, 25% VL, 25% TS **disjoint sets**



- TR**: *Training set* is used to fit (to minimize R_{emp} in SLT) [**training**]^L
- VL**: *Validation set* (or *selection set*) is used to select the best model (among different models and/or hyper-parameters configurations) [**model selection**]
- TR+VL sometimes are jointly called development/design set, i.e. used to build the final model (training different models and selecting the best final model)
- TS**: *Test set* is used for estimation of generalization error (of the final model) (to estimate R in SLT) [**model assessment**]

Notes:

1) the estimation made for model selection (on VL set) [is for model selection purpose], it is not a good estimation for the assessment phase/risk test.

2) **Test set results** cannot be used for model selection (or call it validation set)

Test or model selection?

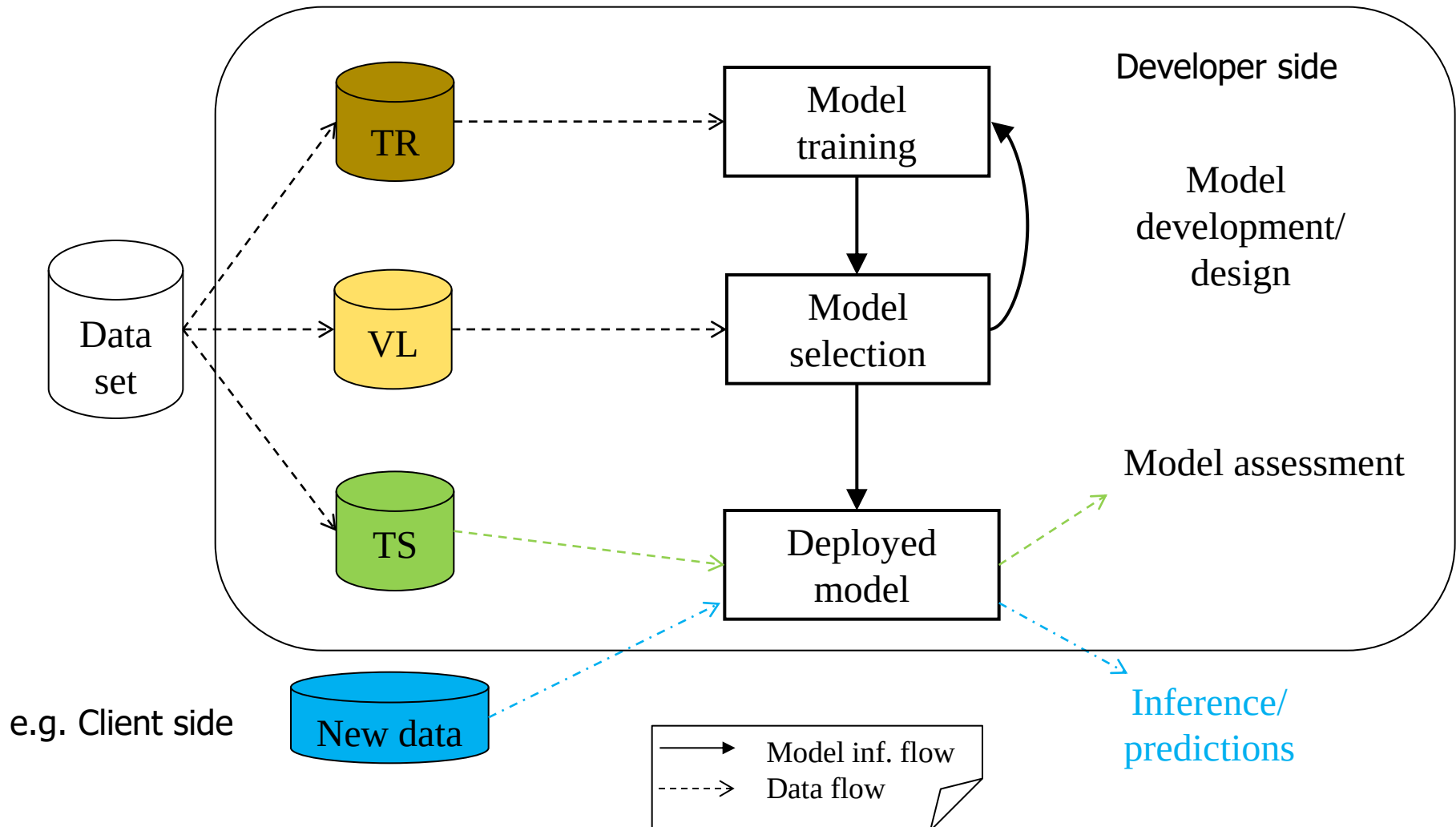
- What if test set is used in a (repeated) design cycle?
 - We are making *model selection* and not reliable *assessment* (estimation of expected generalization error)
 - and *we wouldn't be able to do that on future examples*.
 - Blind test set concept (e.g. for ML competitions)
 - Image an exam exercise: if you see the solutions, it is not a test!
 - In that case, used test set error provides an overoptimistic evaluation of the true test error (→ we will see how easy is to obtain very high classification accuracy over random task even using the test set only implicitly)

Gold rule:

Keep separation between goals and use separate sets in each phase
(TR for training, VL for model selection, TS for risk estimation)



TR/VL/TS by a simple schema



Counterexample

(relevance of a correct assessment)

- 20-30 examples, 1000 random value input variables,
- **Random** target 0/1
- We select 1 model with 1 input var/feature that guess (for chance) 99% (or 100%) on any successive TR and VL and TS set splitting

? Perfect result (a model with accuracy 99%)? What is wrong?

99%/100% is not good estimation of the test error (the right one is 50%)

1) Estimation of error on TR and VL is NOT good risk estimation.

2) Using the entire dataset for feature/model selection prejudice the estimation (biased estimation, called *subset selection bias*)

- Test set was implicitly used at the beginning (*).
- Test set must be separated in advance, before making any model selection (even of the Feature selection) !

On external test the result is 50% (random coin result)!!!!

We are discussing the correctness of the estimation, not the possibility to solve the task !

K-fold CV and other techniques does not solve the issue
[but can confuse you ! ;-)]

The table for the Counterexample



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Pattern

Input variable value

	1	2	...	26	27	28	...	1000	Target
1	...	...	...	1	1	1	...	...	1
2				0	0	0			0
3				1	1	1			1
4				0	0	0			0
...				0	0	0			0
...				1	1	1			1
...				0	0	0			0
20	...	...	...	1	1	1	...	...	1
TS1				1	0	0			1
TS2				0	1	0			0
TS3				1	1	1			1
Accuracy				100%	33%	66%			

Grid search (model selection phase)



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- *Hyper-parameters*. i.e. parameters that are not directly learnt within estimators, can be set by searching a *hyper-parameters* space (of course by model selection on a *validation set*). Search *best* hyper-parameter values.
- **Exhaustive Grid Search**
exhaustively generates candidates from a grid of parameter values,

TRIVIAL EXAMPLE:

Hyper-param.	Lambda 0.1	Lambda 0.01	Lambda 0.001
1 unit	Res1	Res4	Res7
10 units	Res2	Res5	Res8
100 units	Res3	Res6	Res9

Example: The best one is **Res3** →
(Units=100, lambda=0.1) is the winner

AUTOMATIZE IT!!!
Parallelization is easy
(independence of trials)

- Alternatives: Randomized Parameter Optimization (*) → see later
- And many others (also in current ML libraries) → see later

Grid search -2

- The cost of search can be high: Cartesian product between the sets of values for each hyper-parameter
- $\#(\text{values in the range})^{\# \text{hyperparameters}}$:
 - e.g. before $3 \times 3 = 3^2 = 9$
 - What with 5-6 hyperpar., each with 10 possible different values?

Hyper-param.	Lambda 0.1	Lambda 0.01	Lambda 0.001
1 unit	Res1	Res4	Res7
10 units	Res2	Res5	Res8
100 units	Res3	Res6	Res9

- Can be useful to fix some hyper-parameters values in a preliminary experimental phase (since they show to be not relevant) [but take care, see FAQ later on seq. selection error]

Two (or more) levels of **nested grid search**:



- Apply a first *coarse grid* search to the hyperparameters with a table on the combinations of **all the possible** (e.g. growing exponential) values too find good intervals (regions)
- Then finer grid-searches can be performed (over smaller and smaller intervals and with selected hyper-parameters) still using **all** the significant hyper-par.

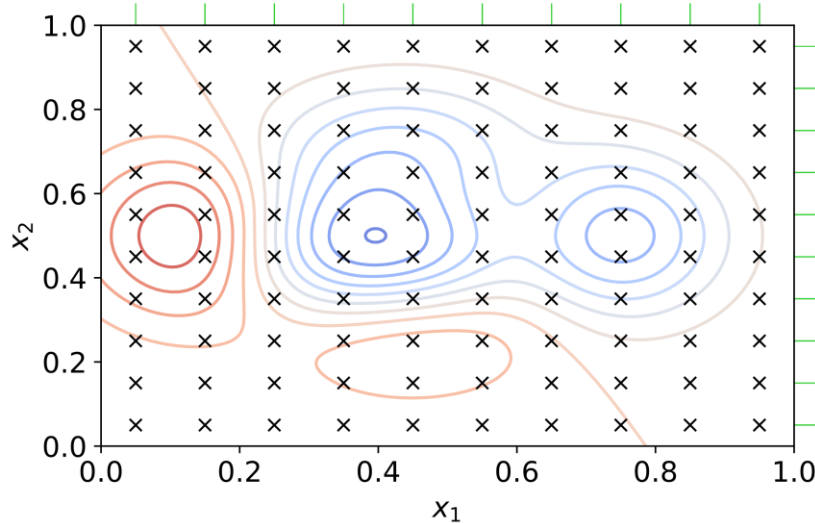
Alternatives to grid search



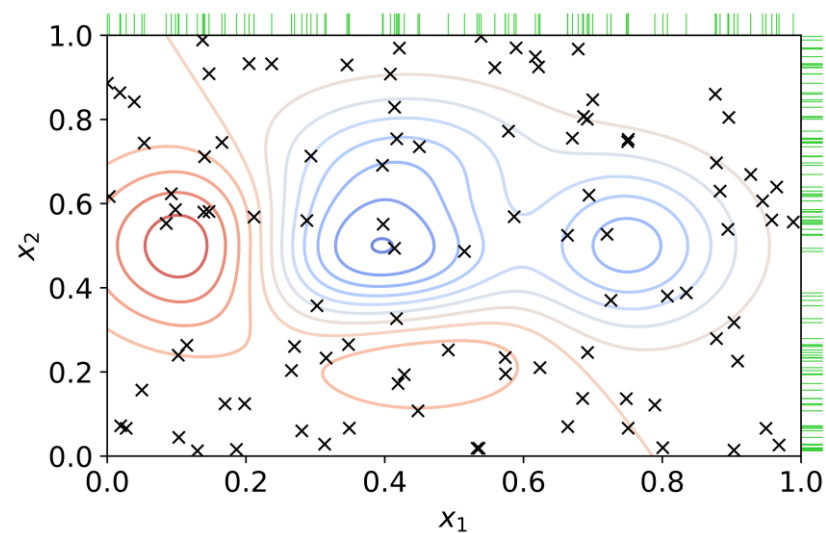
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- Grid search is very useful to observe the behavior of the model changing the hyper-parameters values (to explore the hyper-parameter space → very important for your understanding)
 - But it is *computationally demanding* for large spaces and does not scale well with the increase in number of hyperparameters
- Alternatives exist, trying:
 - to reduce the cost: e.g. *random search*, which avoid to resample the same values of influent hyperparameter (in case some other are influent), and allows to fix the budget of samples independently from the # of hyper-par. See Bergstra, Y. Bengio, (2012). "Random Search for Hyper-Parameter Optimization" J. Machine Learning Research 13: 281–305.
 - to automatize the search: e.g. *Bayesian approaches, evolutionary approaches, configuration evaluation methods* (e.g. Hyperband*), et al.
- Current libraries, as Keras tuner for NN and Scikit-Learn, include various alternatives (toward *AutoML*) → anyway, don't try them uncritically, compare with the grid search that teaches you more for the first approach to ML

Grid versus Random search



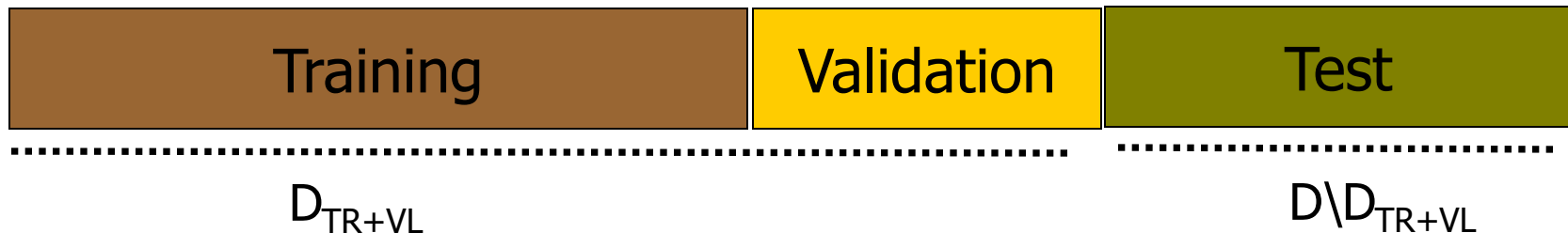
Grid search for 2 hyperparameters:
a total of 100 (10x10) different combinations are evaluated and compared. **Blue** contours indicate regions with strong results, whereas **red** ones show regions with poor results.



Random search for 2 hyperparameters:
100 different random choices are evaluated. The green bars show that more individual values for each hyperparameter are considered compared to a grid search.

Hold out - rethinking

- If data set size is sufficient: [e.g.](#) 50% TR, 25% VL, 25% TS



- How much data is enough?
 - It depends on model complexity, signal-to-noise ratio of the underlying function (e.g. Few for linear model, linear separable task, no noise)
- How to deal with data that are insufficient to make 3 significant/large size splits?
- Can we avoid to be sensible to the particular partitioning of examples?

K-fold CV can help!

Notes and Terminologies

- Generically we have used the term "*cross validation*" for both:
 - An approach to ***model selection*** based on direct estimation of prediction error **by resampling** (instead of analytical)
 - Choose hyperparameters estimating errors
 - More specifically for implementations of estimation methods with resampling specifying how to divide data for both *model selection and assessment*
 - How to make partitions of the data set : e.g. by **K-fold CV**

K-fold CV recall

K-fold Cross-Validation



Fold 4

- Split the data set D into k mutually exclusive subsets $D_1, D_2, \dots, D_K$
- Train the learning algorithm on $D \setminus D_i$ and test it on D_i
- *It uses all the data for training and testing*
- You don't get an unique model (see next lect.) but
- You get also a **variance** (standard deviation) over different folders for your **<class of models/alg.>**

It can be used both for the validation set or for the test set

Issues:

- How many folds? 3-fold, 5-fold, 10-fold, ..., 1-leave-out
- Often computationally very expensive
- Combinable with validation set, double-K-fold CV,

Fold-1	Res on D_1
Fold-2	Res on D_2
Fold-3	Res on D_3
...	...
TOTAL	MEAN +/- SD

An **example** of model selection and assessment with **K-fold CV**



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- Split data in **TR** and **Test** set (here simple hold-out or a K-fold CV)
- **[Model selection]** Use K-fold CV (internal) over **TR** set, obtaining new **TR** e **VL** set in each folder, to find best hyper-parameters of your model (e.g. polynomial order, λ of ridge regression, #units in a NN, ...):
Grid-search with many possible values of the hyper-parameters
 - i.e. for example a k-fold-CV for ($\lambda = 0.1$, #units=20, ...), a k-fold-CV for ($\lambda = 0.01$ #units=20), ... i.e. k-fold-CV for each cell of the grid matrix, ... and then take the best configuration (λ , #units, ...)
(comparing the mean error computed over the validation sets obtained by the all the folds of each k-fold CV, i.e. the result on the “diagonal”)
- Train on the *whole* (original) **TR** set the final model
- **[Model assessment]** Evaluate it on the external **Test** set
 - Question: Can we retrain the model again with all the data?

We will see more combinations in the next lecture!

Before...

- Before enter in general method for model selection/assessment we face (in brief) some particular cases



Lucky/Unlucky sampling

- Can we avoid to be sensible to the particular partitioning of examples? (bias of the particular sample)
- Stratification procedure:
 - *Stratification* is the process of grouping members of the population into relatively homogeneous subgroups before sampling.
- Classification: for each (random) partition (TR and TS in hold-out/CV) each class is represented in approximately the same proportions as in the full data set
- Folder composition: check if order in data related to specific meaning
- Repeated hold-out method or CV: repeat splitting with different random sampling
 - E.g. repeat the CV 10 times !
 - Average the results to yield an overall estimation
 - Useful especially for comparing different models !

Very few data (informal)



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- With very few data: difficult to say whether a sample is representative or not
- Stratification
- Avoid (or consider in evaluation):
 - Missing classes or features in training data
 - Special classes of data not sampled in TR or TS (again stratification)
 - Prior known outliers can affect the mean test results
 - On the opposite: selection of only “easy” cases
- Also: blind test set can be misleading if it is
 - From a different distribution
 - Measured with different scale, different tolerance, etc.
 - Uncleaned, unpreprocessed, ...
 - Extrapolation (out of range data)

Error Functions for Evaluation

- Measuring error (addenda to slides in the intro)
- Classification: see first lessons:
 - **Accuracy** (correct classification rate/error rate often in %), confusion matrix (specificity, sensitivity, ROC curve)
- Regression: residue $r_i = (y_i - o_i)$ $y = target$
 - **MS error** ($mean_i [r^2]$) → **RMS** (root mean square) or S
 - **Mean absolute error** (mean of the absolute error $|r|$)
 - **Max absolute error** ($max_i [|r_i|]$)
 - **R** (correlation coefficient/index) (R^2 coeff. of determination):
E.g. $R = \sqrt{1 - (S^2/S_y^2)}$ where $S_y^2 = mean_i [(y_i - mean_i [y])^2]$ (variance of y)
 - degree of linear dependence between the variables y e o in $[0-1]$, where 1 is the best
 - **Outputs versus targets plots**
 - Various **statistical tests** and statistical significance analysis

Other approaches



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- Bootstrap: random resampling with replacement and repeat it to form different subsets (vl or ts)

REPETITA: 2 gold rules

- TR result is not a good estimation of VL and TS result
- VL result is not a good estimation of TS result

Hence:

- Do **not** use validation set results for test estimation [*risk estimation/model assessment*]
- Do **not** use test set results for *model selection* (in any form*!!!) (or call it validation set)

Or even **1 rule**:

Keep separation between goals and use separate sets in each phase



In the following

- Correct use of cross-validation methods for model selection and/or estimation of risk (assessment)
- Simple concepts with a rich formal notations, can help to avoid ambiguities:
- Very useful guide. It is not intended as a recipe book but help you to rationalize, to understand in a systematic form issues of a rigorous approach.
- Then you have to be *reasonable (when you are aware)*
- In any case use them if you have uncertainties, it is better than to use *fantasy!*

Exercise (in the slide of the next lecture):

relate wrong cases 1 and 2 of slide "counterexample" to misuses of such schemas!

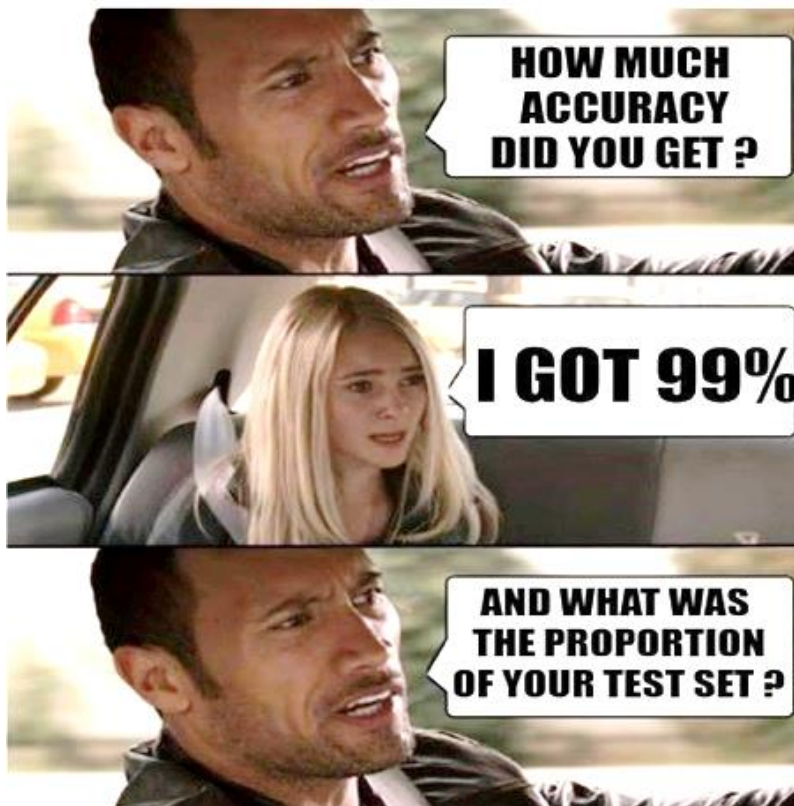
Bibliography

- Hastie, Tibshirani, Friedman: Chapter 7
- Duda, Hart, Stork: 9.4, 9.6
- Cherkassky, Mulier: 3.4.2
- Haykin: e.g. 4.14 2nd edition - 4.13 3rd edition
- Mitchell: e.g. 4.6.5, 5.1, 5.6
- Note that often the space in such books is small with respect to the relevance of the topic!!!

For fun

- Can I have just a look to the test set?

See <https://youtu.be/XvOsh15hLIIs>



By Sepe-Dukic past students



* It can hold also for the validation set used as test set ;-)